

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-106660295})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 3459 & 2637 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}3459 + \mathbf{Z}(2637 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 26665074 = 0$, of discriminant -106660295 . Class group invariants $(110 \ 10 \ 10)$, class number 11000; the three stored generator ideals (HNF) are $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 192 & 149 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 104 & 101 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1881y^3 + (36s - 18)y^2 - 2214820y + (-3550s + 1775)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 3762y^8 - 891479y^6 + 26225782740y^4 - 1910165218100y^2 + 336046591934375$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ -2 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{628074433}{495168272762461299} + \left(\frac{61462}{495168272762461299}\right)s, \quad J = \begin{pmatrix} 3459 & 2637 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3459$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $21808248011093 = 757 \cdot 809 \cdot 35610361$:

$$\begin{pmatrix} 21808248011093 & 5834823254305 & 21133629511890 & 10464580457876 & 4163658108515 & 6680979443504 & 400281690000 & 19658449438256 & 18320021830249 & 1847010098211 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (0 \ 1 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1881y^3 + (36s - 18)y^2 - 2214820y + (-3550s + 1775)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 3762y^8 - 891479y^6 + 26225782740y^4 - 1910165218100y^2 + 336046591934375$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ -2 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7063831}{103621065612199888} + \left(\frac{88117}{207242131224399776}\right)s, \quad J = \begin{pmatrix} 2906 & 1111 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2906$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $48535752789461 = 7 \cdot 17 \cdot 359 \cdot 1136109941$:

$$\begin{pmatrix} 48535752789461 & 20684831812230 & 41919225679231 & 22233634708063 & 432666772455 & 34560384028454 & 18999763034252 & 234980359461 & 43608127528486 & 8578710155817 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 1 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1881y^3 + (36s - 18)y^2 - 2214820y + (-3550s + 1775)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 3762y^8 - 891479y^6 + 26225782740y^4 - 1910165218100y^2 + 336046591934375$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ -2 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{209517721}{100421421980737824} + \left(-\frac{46037}{100421421980737824}\right)s, \quad J = \begin{pmatrix} 2514 & 1788 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2514$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $43529656246177 = 17 \cdot 89 \cdot 28770427129$:

$$\begin{pmatrix} 43529656246177 & 23760474300758 & 26977823591851 & 25977725701912 & 7835074577365 & 21096205842733 & 29843910486809 & 7739060751154 & 21452112775666 & 8217485668397 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 877)y^3 + (-12s - 96416)y^2 + (-415s + 375030)y + (-1650s + 11227340)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 1749y^8 - 189338y^7 + 28568659y^6 - 449980264y^5 \\ & - 9697008411y^4 + 85627040190y^3 + 3623855094940y^2 + 44933719337 \\ & 800y + 198630302329600 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ -1 \ 0 \ -1 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{628074433}{495168272762461299} + \left(\frac{61462}{495168272762461299}\right)s, \quad J = \begin{pmatrix} 3459 & 2637 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3459$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3864710714368 = 2^{10} \cdot 7^3 \cdot 19 \cdot 761^2$:

$$\begin{pmatrix} 809704 & 681856 & 3052 & 601944 & 597057 & 323858 & 391118 & 694745 & 84968 & 167325 \\ 0 & 21308 & 19264 & 21280 & 11396 & 6240 & 21075 & 20850 & 57 & 13697 \\ 0 & 0 & 4 & 0 & 0 & 0 & 2 & 1 & 3 & 2 \\ 0 & 0 & 0 & 28 & 19 & 2 & 3 & 24 & 9 & 14 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 117 factors with signed exponents of absolute value up to 237209; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 877)y^3 + (-12s - 96416)y^2 + (-415s + 375030)y + (-1650s + 11227340)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 1749y^8 - 189338y^7 + 28568659y^6 - 449980264y^5 \\ & - 9697008411y^4 + 85627040190y^3 + 3623855094940y^2 + 44933719337 \\ & 800y + 198630302329600 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ -1 \ 0 \ -1 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7063831}{103621065612199888} + \left(\frac{88117}{207242131224399776}\right)s, \quad J = \begin{pmatrix} 2906 & 1111 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2906$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $52061859799040 = 2^{13} \cdot 5 \cdot 7 \cdot 13 \cdot 17 \cdot 19 \cdot 83 \cdot 521$:

$$\begin{pmatrix} 50841659960 & 12130262112 & 37652619184 & 26851228606 & 47114686780 & 36745508632 & 17341787248 & 39892124230 & 40486669015 & 7624246489 \\ 0 & 8 & 4 & 4 & 1 & 0 & 4 & 6 & 1 & 7 \\ 0 & 0 & 4 & 2 & 2 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 4 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 4 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 120 factors with signed exponents of absolute value up to 358157; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 877)y^3 + (-12s - 96416)y^2 + (-415s + 375030)y + (-1650s + 11227340)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 1749y^8 - 189338y^7 + 28568659y^6 - 449980264y^5 \\ & - 9697008411y^4 + 85627040190y^3 + 3623855094940y^2 + 44933719337 \\ & 800y + 198630302329600 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ -1 \ 0 \ -1 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{209517721}{100421421980737824} + \left(-\frac{46037}{100421421980737824}\right)s, \quad J = \begin{pmatrix} 2514 & 1788 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2514$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $540412579840 = 2^{12} \cdot 5 \cdot 7^3 \cdot 19 \cdot 4049$:

$$\begin{pmatrix} 603139040 & 265790784 & 213675672 & 595758800 & 106712749 & 211161735 & 542220938 & 219612094 & 369168465 & 269438005 \\ 0 & 16 & 0 & 13 & 1 & 10 & 6 & 6 & 14 & 7 \\ 0 & 0 & 56 & 20 & 24 & 50 & 55 & 35 & 13 & 23 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 126 factors with signed exponents of absolute value up to 569085; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 1510)y^3 + (-23s + 41674)y^2 + (-176s + 1000416)y + (-700s + 7335720)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 3020y^8 + 86346y^7 + 30864015y^6 + 1113401064y^5 \\ & + 22191032252y^4 + 314403718160y^3 + 3296468512064y^2 + 212458261 \\ & 74720y + 66873539174400 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 1 \ 0 \ 0 \ -1 \ 0 \ -1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{628074433}{495168272762461299} + \left(\frac{61462}{495168272762461299}\right)s, \quad J = \begin{pmatrix} 3459 & 2637 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3459$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $93791761116912 = 2^4 \cdot 3^8 \cdot 271 \cdot 3296897$:

$$\begin{pmatrix} 96493581396 & 18390163599 & 44798973900 & 31195566558 & 36712657159 & 58877142214 & 26863201002 & 88581604925 & 1910607782 & 86381622132 \\ 0 & 3 & 0 & 0 & 2 & 0 & 2 & 2 & 1 & 1 \\ 0 & 0 & 3 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 54 & 53 & 39 & 50 & 15 & 21 & 23 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 93 factors with signed exponents of absolute value up to 53323760403; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (1 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 1510)y^3 + (-23s + 41674)y^2 + (-176s + 1000416)y + (-700s + 7335720)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 3020y^8 + 86346y^7 + 30864015y^6 + 1113401064y^5 \\ & + 22191032252y^4 + 314403718160y^3 + 3296468512064y^2 + 212458261 \\ & 74720y + 66873539174400 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 1 \ 0 \ 0 \ -1 \ 0 \ -1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7063831}{103621065612199888} + \left(\frac{88117}{207242131224399776}\right)s, \quad J = \begin{pmatrix} 2906 & 1111 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2906$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $240768953510400 = 2^9 \cdot 3^2 \cdot 5^2 \cdot 7 \cdot 1453 \cdot 205487$:

$$\begin{pmatrix} 250800993240 & 1372411147000 & 3579948406 & 248832458530 & 86257672909 & 137710451145 & 218540851808 & 130407476222 & 208095132380 & 152103259492 \\ 0 & 60 & 30 & 20 & 23 & 15 & 9 & 50 & 7 & 7 \\ 0 & 0 & 2 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 93 factors with signed exponents of absolute value up to 74800076876; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (1 \ 3 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 1510)y^3 + (-23s + 41674)y^2 + (-176s + 1000416)y + (-700s + 7335720)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 3020y^8 + 86346y^7 + 30864015y^6 + 1113401064y^5 \\ & + 22191032252y^4 + 314403718160y^3 + 3296468512064y^2 + 212458261 \\ & 74720y + 66873539174400 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 1 \ 0 \ 0 \ -1 \ 0 \ -1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{209517721}{100421421980737824} + \left(-\frac{46037}{100421421980737824}\right)s, \quad J = \begin{pmatrix} 2514 & 1788 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2514$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $81961158327552 = 2^8 \cdot 3^4 \cdot 7 \cdot 1019 \cdot 554129$:

$$\begin{pmatrix} 47431225884 & 15771391968 & 35485407492 & 38537250028 & 38260024804 & 41561872238 & 31478083928 & 42114838365 & 43918366874 & 47030527372 \\ 0 & 12 & 0 & 4 & 7 & 4 & 8 & 8 & 5 & 6 \\ 0 & 0 & 12 & 4 & 10 & 11 & 6 & 7 & 8 & 0 \\ 0 & 0 & 0 & 4 & 1 & 3 & 3 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 & 3 & 0 & 2 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 89 factors with signed exponents of absolute value up to 86121435080; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \ 3 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 1221)y^3 + (-39s - 82872)y^2 + (-1008s + 1676781)y + (40122s + 132676299)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 2440y^8 - 163342y^7 + 31673031y^6 - 1615497435y^5 \\ & - 10685395611y^4 + 3634416891864y^3 - 75544249516236y^2 - 171195 \\ & 8162915520y + 60533089960007295 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 0 \ -1 \ -1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{628074433}{495168272762461299} + \left(\frac{61462}{495168272762461299}\right)s, \quad J = \begin{pmatrix} 3459 & 2637 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3459$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $113744432395125 = 3^3 \cdot 5^3 \cdot 7 \cdot 4814579149$:

$$\begin{pmatrix} 505530810645 & 463941284880 & 267728442435 & 54969227505 & 443218255004 & 108726571234 & 415291919916 & 426573618015 & 291113101741 & 323418708103 \\ 0 & 15 & 0 & 0 & 1 & 11 & 0 & 14 & 8 & 9 \\ 0 & 0 & 5 & 2 & 0 & 4 & 4 & 0 & 4 & 3 \\ 0 & 0 & 0 & 3 & 2 & 1 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 331 factors with signed exponents of absolute value up to 39114723232309; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 4 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 1221)y^3 + (-39s - 82872)y^2 + (-1008s + 1676781)y + (40122s + 132676299)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 - 2440y^8 - 163342y^7 + 31673031y^6 - 1615497435y^5 \\ - 10685395611y^4 + 3634416891864y^3 - 75544249516236y^2 - 171195 \\ 8162915520y + 60533089960007295$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 0 \ -1 \ -1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7063831}{103621065612199888} + \left(\frac{88117}{207242131224399776}\right)s, \quad J = \begin{pmatrix} 2906 & 1111 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2906$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $683444964983985 = 3^2 \cdot 5 \cdot 1518766588533$:

$$\begin{pmatrix} 227814988327995 & 13495041524460 & 130174409700515 & 212468347312785 & 219157139311409 & 20034470865309 & 209317044922506 & 89746304750640 & 113370583763036 & 167381246951893 \\ 0 & 3 & 1 & 0 & 1 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 334 factors with signed exponents of absolute value up to 144565173857909; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 2 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 1221)y^3 + (-39s - 82872)y^2 + (-1008s + 1676781)y + (40122s + 132676299)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 - 2440y^8 - 163342y^7 + 31673031y^6 - 1615497435y^5 \\ - 10685395611y^4 + 3634416891864y^3 - 75544249516236y^2 - 171195 \\ 8162915520y + 60533089960007295$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 0 \ -1 \ -1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{209517721}{100421421980737824} + \left(-\frac{46037}{100421421980737824}\right) s, \quad J = \begin{pmatrix} 2514 & 1788 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2514$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $87534615181419 = 3^5 \cdot 7 \cdot 23971 \cdot 2146789$:

$$\begin{pmatrix} 3242022784497 & 3120551347569 & 1781573678223 & 768605153481 & 452115395216 & 220472116784 & 1109362670874 & 2253096761205 & 1539409041683 & 2688253619641 \\ 0 & 3 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 2 & 2 & 2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 339 factors with signed exponents of absolute value up to 218099500408625; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (s + 1797)y^3 + (-16s - 219447)y^2 + (-263s + 2983268)y + (2546s - 7929745)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 + 3604y^8 - 449695y^7 + 37179083y^6 - 1675978848y^5 \\ & + 51732873662y^4 - 977639481840y^3 + 12050928620806y^2 - 8301318 \\ & 8530561y + 235706769450839 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ -1 \ 0 \ -1 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{628074433}{495168272762461299} + \left(\frac{61462}{495168272762461299}\right) s, \quad J = \begin{pmatrix} 3459 & 2637 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3459$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1346413176596005 = 5 \cdot 7 \cdot 7159 \cdot 5373508577$:

$$\begin{pmatrix} 1346413176596005 & 550005499012546 & 913638317690649 & 690599832420529 & 1101271366357053 & 148222726627857 & 876544696714855 & 302809864642500 & 502024132418327 & 63876572328771 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 386 factors with signed exponents of absolute value up to 6995885778952; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 3 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (s + 1797)y^3 + (-16s - 219447)y^2 + (-263s + 2983268)y + (2546s - 7929745)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 + 3604y^8 - 449695y^7 + 37179083y^6 - 1675978848y^5 \\ & + 51732873662y^4 - 977639481840y^3 + 12050928620806y^2 - 8301318 \\ & 8530561y + 235706769450839 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ -1 \ 0 \ -1 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7063831}{103621065612199888} + \left(\frac{88117}{207242131224399776}\right)s, \quad J = \begin{pmatrix} 2906 & 1111 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2906$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $113431576876625 = 5^3 \cdot 7 \cdot 17 \cdot 6007 \cdot 1269461$:

$$\begin{pmatrix} 4537263075065 & 3508820104315 & 4457686810690 & 1082990814211 & 4455782888743 & 199314049086 & 777157767647 & 1014504016260 & 2725283420073 & 2764282881801 \\ 0 & 5 & 0 & 1 & 1 & 4 & 1 & 0 & 1 & 3 \\ 0 & 0 & 5 & 4 & 1 & 0 & 4 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 381 factors with signed exponents of absolute value up to 8453124904169; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (s + 1797)y^3 + (-16s - 219447)y^2 + (-263s + 2983268)y + (2546s - 7929745)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 + 3604y^8 - 449695y^7 + 37179083y^6 - 1675978848y^5 \\ & + 51732873662y^4 - 977639481840y^3 + 12050928620806y^2 - 8301318 \\ & 8530561y + 235706769450839 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ -1 \ 0 \ -1 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{209517721}{100421421980737824} + \left(-\frac{46037}{100421421980737824}\right)s, \quad J = \begin{pmatrix} 2514 & 1788 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2514$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $151454047292905 = 5 \cdot 7^2 \cdot 17 \cdot 19 \cdot 107 \cdot 17886629$:

$$\begin{pmatrix} 21636292470415 & 2777737321912 & 9444215882468 & 7140550998799 & 20815017408200 & 2985918302308 & 6557805212005 & 18173701127952 & 7781202680913 & 8272009911399 \\ 0 & 7 & 0 & 6 & 1 & 6 & 6 & 2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 381 factors with signed exponents of absolute value up to 11184926418494; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \ 3 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s + 3222)y^3 + (20s + 282696)y^2 + (1396s - 2087160)y + (-656s - 87692192)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 + 6459y^8 + 539640y^7 + 30608564y^6 + 596178464y^5 \\ & + 3398291520y^4 - 220751263744y^3 + 6036551389184y^2 + 3170957521 \\ & 71520y + 7701453005127680 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ -1 \ 1 \ 1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{628074433}{495168272762461299} + \left(\frac{61462}{495168272762461299}\right)s, \quad J = \begin{pmatrix} 3459 & 2637 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3459$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $267124184985088 = 2^9 \cdot 17 \cdot 113^2 \cdot 2403463$:

$$\begin{pmatrix} 36936419384 & 17308250680 & 32550810830 & 5215878061 & 3679901483 & 12167536014 & 33412165383 & 25579986456 & 33895112601 & 21296870524 \\ 0 & 452 & 0 & 450 & 37 & 0 & 180 & 134 & 294 & 3 \\ 0 & 0 & 2 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 197 factors with signed exponents of absolute value up to 614977045; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s + 3222)y^3 + (20s + 282696)y^2 + (1396s - 2087160)y + (-656s - 87692192)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 + 6459y^8 + 539640y^7 + 30608564y^6 + 596178464y^5 \\ & + 3398291520y^4 - 220751263744y^3 + 6036551389184y^2 + 3170957521 \\ & 71520y + 7701453005127680 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ -1 \ 1 \ 1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7063831}{103621065612199888} + \left(\frac{88117}{207242131224399776}\right)s, \quad J = \begin{pmatrix} 2906 & 1111 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2906$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $481904011282900 = 2^2 \cdot 5^2 \cdot 577 \cdot 8351889277$:

$$\begin{pmatrix} 48190401128290 & 15089846386290 & 2593477230179 & 40203499376119 & 25081025194083 & 37609245131089 & 40148273987758 & 30271870955066 & 3734929388947 & 12233685325432 \\ 0 & 10 & 6 & 2 & 3 & 6 & 8 & 5 & 0 & 5 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 195 factors with signed exponents of absolute value up to 1650760; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s + 3222)y^3 + (20s + 282696)y^2 + (1396s - 2087160)y + (-656s - 87692192)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 + 6459y^8 + 539640y^7 + 30608564y^6 + 596178464y^5 \\ & + 3398291520y^4 - 220751263744y^3 + 6036551389184y^2 + 3170957521 \\ & 71520y + 7701453005127680 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ -1 \ 1 \ 1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{209517721}{100421421980737824} + \left(-\frac{46037}{100421421980737824}\right)s, \quad J = \begin{pmatrix} 2514 & 1788 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2514$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $515077082455040 = 2^{10} \cdot 5 \cdot 7^2 \cdot 17 \cdot 43 \cdot 2808593$:

$$\begin{pmatrix} 4598902521920 & 894729359118 & 3516780975776 & 2415140113855 & 2982561655812 & 4037883890922 & 934544403959 & 3673579685834 & 314779591982 & 4296125929399 \\ 0 & 14 & 6 & 4 & 4 & 4 & 11 & 8 & 5 & 4 \\ 0 & 0 & 2 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 220 factors with signed exponents of absolute value up to 165798006; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.